

Assignment - 2

1. What are the different protocols you observe at the following layers of the protocol stack?

Application layer	Transport Layer	Network Layer
NBNS	UDP	ARP
DHCP	TCP	ICMPv6
DNS		
MDNS		
HTTP		

2. What is the total amount of data being received for the following two cases?

A) When you access: <http://neverssl.com/>

B) When you access: <http://www.cornell.edu/>

Jul 9 18:59

*wlp0s20f3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help



ip.addr==10.85.17.213

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
311	10.039759093	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
595	20.080395579	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
894	30.117667434	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
1194	40.053078070	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
1535	50.088810567	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
1847	60.124205660	10.85.17.213	255.255.255.255	UDP	868	59648 → 29810 Len=826
2150	70.161190018	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
2458	80.197372452	10.85.17.213	255.255.255.255	UDP	867	59648 → 29810 Len=825
2753	90.235732550	10.85.17.213	255.255.255.255	UDP	867	57825 → 29810 Len=825
3034	100.272471278	10.85.17.213	255.255.255.255	UDP	867	57825 → 29810 Len=825
3343	110.309545636	10.85.17.213	255.255.255.255	UDP	868	57825 → 29810 Len=826
3676	120.345318210	10.85.17.213	255.255.255.255	UDP	867	57825 → 29810 Len=825
4027	130.381516572	10.85.17.213	255.255.255.255	UDP	868	57825 → 29810 Len=826

Frame 1: 867 bytes on wire (6936 bits), 867 bytes captured (6936 bits) on interface wlp0s20f3, id 0
Ethernet II, Src: b0:19:21:74:06:0c (b0:19:21:74:06:0c), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
Internet Protocol Version 4, Src: 10.85.17.213, Dst: 255.255.255.255
User Datagram Protocol, Src Port: 59648, Dst Port: 29810
Data (825 bytes)

```
0000 ff ff ff ff ff ff b0 19 21 74 06 0c 08 00 45 a0 .....!t...E-
0010 03 55 6f 2f 00 00 40 11 eb 9f 0a 55 11 d5 ff ff ...Uo/-@...U....
0020 ff ff e9 00 74 72 03 41 55 f9 00 00 03 35 7b 22 ....tr.AU....5{"
0030 68 65 61 64 65 72 22 3a 7b 22 76 65 72 73 69 6f header":{"versio
0040 6e 22 3a 22 31 2e 33 2e 30 22 2c 22 6d 61 63 22 n":"1.3. 0","mac"
0050 3a 22 42 30 3a 31 39 3a 32 31 3a 37 34 3a 30 36 :B0:19: 21:74:06
0060 3a 30 43 22 2c 22 74 79 70 65 22 3a 31 2c 22 64 :0C","ty pe":1,"d
0070 65 76 69 63 65 22 3a 22 61 70 22 2c 22 65 72 72 evice":" ap","err
0080 6f 72 22 3a 30 2c 22 64 65 73 74 22 3a 22 30 31 or":0,"d est":"01
0090 66 66 61 64 66 34 32 34 33 33 31 34 36 66 62 61 ffadf424 33146fba
00a0 65 34 66 38 63 37 39 35 63 33 62 63 63 37 22 2c e4f8c795 c3bcc7",
00b0 22 76 65 72 43 61 70 22 3a 33 70 2c 22 62 6f 64 "verCap" :3},"bod
00c0 79 22 3a 7b 22 64 65 76 09 63 65 49 6e 66 6f 22 y":{"dev iceInfo"
00d0 3a 7b 22 6d 6f 64 65 6c 22 3a 22 45 41 50 32 32 :{"model ": "EAP22
00e0 35 22 2c 22 6e 61 6d 65 22 3a 22 45 41 50 32 32 5","name ": "EAP22
00f0 35 2d 42 30 2d 31 39 2d 32 31 2d 37 34 2d 30 36 5-B0-19- 21-74-06
0100 2d 30 43 22 2c 22 66 69 72 6d 77 61 72 65 56 65 -0C","fi rmwareVe
0110 72 73 69 6f 6e 22 3a 22 35 2e 31 2e 35 20 42 75 rsion":" 5.1.5 Bu
0120 69 6c 64 20 32 30 32 34 30 33 31 33 20 52 65 6c ild 2024 0313 Rel
0130 2e 20 33 33 39 32 38 22 2c 22 6d 6f 64 65 6c 56 . 33928","modelV
0140 65 72 73 69 6f 6e 22 3a 22 34 2e 30 22 2c 22 68 ersion": "4.0","h
0150 61 72 64 77 61 72 65 56 65 72 73 69 6f 6e 22 3a ardwareV ersion":
0160 22 34 2e 30 22 2c 22 75 70 54 69 6d 65 22 3a 22 "4.0","u pTime":
0170 31 36 20 64 61 79 73 20 30 33 3a 34 33 3a 33 36 16 days 03:43:36
0180 22 2c 22 63 70 75 55 74 09 22 3a 31 2c 22 6d 65 ", "cpuUt i":1,"me
0190 6d 55 74 69 22 3a 35 32 2c 22 74 78 52 61 74 65 mUti":52, "txRate
01a0 22 3a 32 30 31 30 2c 22 72 78 52 61 74 65 22 3a ":2010," rxRate":
01b0 37 37 37 36 2c 22 77 69 72 65 6c 65 73 4c 69 7776,"wi relessLi
01c0 6e 6b 65 64 22 3a 66 61 6c 73 65 2c 22 69 73 46 nked":fa lse,"isF
```

wireshark_wlp0s20f358D3R3.pcapng

Packets: 4296 - Displayed: 14 (0.3%) - Marked: 1 (0.0%) - Dropped: 0 (0.0%) Profile: Default

ip.addr==10.85.23.127					
No.	Time	Source	Destination	Protocol	Length Info
2	0.000000593	10.85.23.127	224.0.0.251	MDNS	119 Standard query 0x0002 PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.local, "QM" question PTR ...
30	0.921492891	10.85.23.127	224.0.0.251	MDNS	119 Standard query 0x0003 PTR _googlecast._tcp.local, "QM" question PTR _674A0243._sub._googlecast._tcp.local, "QM" question PTR ...
594	19.762621979	10.85.23.127	224.0.0.251	MDNS	82 Standard query 0x0001 PTR _googlecast._tcp.local, "QU" question

3. Answer the following when you access the site
<http://neverssl.com/>

(A) How many HTTP GET requests do you observe?
List down the GET requests.

→ One HTTP GET requests observed.

GET requests:

Time	Source	Destination	Protocol	length
30.631015453	10.85.17.18	34.223.124.45	HTTP	142

Info

GET/HTTP/1.1

(B) What version of HTTP the server is running?

GET/HTTP/1.1

httprequestmethod=GET

No.	Time	Source	Destination	Protocol	Length	Info
924	30.631015453	10.85.17.18	34.223.124.45	HTTP	142	GET / HTTP/1.1

Protocol

(C) For each of the HTTP Get requests, find out:

i) The total number of TCP segments being received.

→ One TCP segment received

ii) The total amount of data being received in the corresponding HTTP response message.

→ 4261 bytes.

No.	Time	Source	Destination	Protocol	Length	Info
914	30.324086695	b0:19:21:74:0d:48	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.22.105
915	30.324086801	b0:19:21:30:c0:de	Broadcast	ARP	60	Who has 10.85.22.103? Tell 10.85.16.223
916	30.324086911	b0:19:21:74:0c:88	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.22.10
917	30.425182592	ASIXElectron_35:2d:...	Broadcast	ARP	60	Who has 10.85.18.15? Tell 10.85.16.1
918	30.425183899	b0:19:21:d4:b8:20	Broadcast	ARP	60	Who has 10.85.22.103? Tell 10.85.17.215
919	30.629102482	10.85.16.36	255.255.255.255	UDP	868	55445 → 29810 Len=826
920	30.629112778	a8:0e:84:cb:36:c8	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.16.36
921	30.629113241	b0:19:21:74:06:14	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.17.220
922	30.630835731	34.223.124.45	10.85.17.18	TCP	74	80 → 41360 [SYN, ACK] Seq=0 Ack=1 Win=26847 Len=0 MSS=1460 SACK_PERM TSval=3443542074 TSecr=1545302119 WS=128
923	30.630879735	10.85.17.18	34.223.124.45	TCP	66	41360 → 80 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=1545302467 TSecr=3443542074
924	30.631015453	10.85.17.18	34.223.124.45	HTTP	142	GET / HTTP/1.1
925	30.735667236	10.85.17.123	255.255.255.255	UDP	866	49090 → 29810 Len=824
926	30.735722955	b0:19:21:d4:90:a6	Broadcast	ARP	60	Who has 10.85.22.156? Tell 10.85.17.123
927	30.735724049	ASIXElectron_35:2d:...	Broadcast	ARP	60	Who has 10.85.19.98? Tell 10.85.16.1
928	30.835506594	10.85.19.55	255.255.255.255	UDP	506	44968 → 29810 Len=464
929	30.935712235	10.85.19.34	255.255.255.255	UDP	506	37558 → 29810 Len=464
930	30.939402538	34.223.124.45	10.85.17.18	TCP	66	80 → 41360 [ACK] Seq=1 Ack=77 Win=29312 Len=0 TSval=3443542471 TSecr=1545302467
931	30.939403013	34.223.124.45	10.85.17.18	HTTP	4327	HTTP/1.1 200 OK (text/html)
932	30.939495836	10.85.17.18	34.223.124.45	TCP	66	41360 → 80 [ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=1545302776 TSecr=3443542471
933	30.940275457	10.85.17.18	34.223.124.45	TCP	66	41360 → 80 [FIN, ACK] Seq=77 Ack=4262 Win=60416 Len=0 TSval=1545302776 TSecr=3443542471
934	31.043401471	10.85.22.209	255.255.255.255	UDP	869	43260 → 29810 Len=827
935	31.043404646	10.85.21.198	224.0.0.251	MDNS	152	Standard query 0x0024 PTR _googlecast._tcp.local, "QM" question PTR %9E5E7C8F47989526C9BCD95D24084F60B27C5ED._sub._go...

```

> Frame 931: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlp0s20f3, id
> Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: 44:38:e8:60:4d:a2 (44:38:e8:60:4d:a2)
> Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.17.18

```

- Transmission Control Protocol, Src Port: 80, Dst Port: 41360, Seq: 1, Ack: 77, Len: 4261

Source Port: 80

Destination Port: 41360

```
[Stream index: 0]
```

```
↳ [Conversation completeness: Complete, WITH_DATA (31)]
```

[TCP Segment Len: 4261]

Sequence Number: 1 (relative sequence number)

Sequence Number (raw): 3464783344

[Next Sequence Number: 4262 (relative sequence number)]

Acknowledgment Number: 77 (relative ack number)

Acknowledgment number (raw): 3400077786

1000 = Header Length: 32 bytes (8)

```
Flags: 0x018 (PSH, ACK)
```

Window: 229

```
[Calculated window size: 29312]
```

```
[Window size scaling factor: 128]
```

Checksum: 0xcb3e [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps

► [Timestamps]

- ▶ [SEQ/ACK analysis]

TCP payload (4261 bytes)

- Hypertext Transfer Protocol
- Line-based text data: text/html (131 lines)

Transmission Control Protocol (tcp), 32 bytes

```

0020 11 12 00 50 1 90 ce 84 65 f0 ca a9 11 da 80 18 . . P . . . e . . .
0030 00 e5 cb 3e 00 00 01 01 88 0a cd 40 49 c7 5c 1b . . . . . / . . . I . .
0040 71 c3 48 54 54 50 2f 31 2e 31 20 32 30 20 4f q . HTTP / . . 200 .
0050 4b 0d 0a 44 61 74 65 3a 20 54 68 75 2c 20 30 39 K . Date : Thu, 09
0060 20 0a 75 6c 20 32 30 32 36 20 31 33 3a 32 35 3a Jul 02 6 13:25:
0070 30 38 20 47 4d 54 0d 0a 53 65 72 76 65 72 3a 20 08 GMT-- Server:
0080 41 70 61 63 68 65 2f 32 2e 3a 2e 36 36 20 28 29 Apache/2 .4.66 ( )
0090 0d 0a 55 70 67 72 61 64 65 3a 20 68 32 2c 68 32 . Upgrad e: h2,h2
00a0 63 0d 0a 43 6f 6e 6e 65 63 74 69 6f 6e 3a 20 55 c . Conne ction: U
00b0 70 67 72 61 64 65 0d 0a 4c 61 73 74 2d 4d 6f 64 pgraded . Last-Mod
00c0 69 66 69 65 64 3a 20 57 65 64 2c 20 32 39 20 4a ified: Wed, 29 J
00d0 75 6e 20 32 30 32 32 20 30 30 3a 32 33 3a 33 33 un 2022 00:23:33
00e0 20 47 4d 54 0d 0a 45 54 61 67 3a 20 22 66 37 39 GMT . ET ag: "f79
00f0 2d 35 65 32 38 62 32 39 64 33 38 65 39 33 22 0d -5e28b29 d38e93".
0100 0a 41 63 63 65 70 74 2d 52 61 6e 67 65 73 3a 20 . Accept- Ranges:
0110 62 79 74 65 73 0d 0a 43 6f 6e 74 65 6e 74 2d 4c bytes . Content-L
0120 65 6e 67 74 68 3a 20 33 39 36 31 0d 0a 56 61 72 length: 3 961 . Var
0130 79 3a 20 41 63 63 65 70 74 2d 45 6e 63 6f 64 y: Accep t-Encodi
0140 6e 67 0d 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 ng . Cont ent-Type
0150 3a 20 74 65 78 74 2f 68 74 6d 6c 3b 0d 63 68 61 : text/html; cha
0160 72 73 65 74 3d 55 54 46 2d 38 0d 0a 0d 0a 3c 68 rset=UTF -8 . . . ch
0170 74 6d 6c 3e 3a 09 3c 68 65 61 64 3e 0a 09 09 3c tml> <h ead> . . <
0180 74 69 74 6c 65 3e 4e 65 76 65 72 53 53 4c 20 2d title> neVSSL -
0190 20 43 6f 6e 6e 65 63 74 69 6e 67 20 2e 2e 2e 20 Connect ing ...
01a0 3c 2f 74 69 74 6c 65 3e 0a 09 09 3c 73 74 79 6c </title> . . . <styl
01b0 65 3e 0a 09 09 62 6f 64 79 20 7b 0a 09 09 66 e3> . . . bod y { . . . f
01c0 6f 6e 74 2d 66 61 6d 69 6c 79 3a 20 4d 6f 6e 74 ont-fami ly: Mont
01d0 73 65 72 72 61 74 2c 20 68 65 6c 76 65 74 69 63 serrat, helvetic
01e0 61 2c 20 61 72 69 61 6c 2c 20 73 61 6e 73 2d 73 a , arial , sans-s

```

Packets: 4296 · Displayed: 4296 (100.0%) · Marked: 1 (0.0%) · Dropped: 0 (0.0%) Profile: Default

(D) Identify TCP three way handshake used to establish the connection with the <http://neverssl.com>. Specify SYN, SYN-ACK packets and write the sequence number for each of the three accesses.

→ SYN Packet:

- Source IP: 10.85.17.18 (Client)
- Destination IP: 34.223.124.45 (Server)
- Protocol info: 57022 → 80 [SYN]
- Sequence Number: 0

SYN-ACK Packet:

- Source IP: 34.223.124.45 (Server)
- Destination IP: 10.85.17.18 (Client)
- Protocol info: 80 → 57022 [SYN, ACK]
- Sequence Number: 0
(ACK=1)

ACK Packet:

- Source IP: 10.85.17.18 (client)
- Destination IP: 34.223.124.45 (server)
- Sequence Number: 1
(ACK=1)

